

E-Commerce Sales Analysis Report

Project Overview

This report analyzes 5,000 e-commerce transactions to evaluate key business performance across sales revenue, customer behavior, product performance, regional sales, and profitability.

The dataset was cleaned and analyzed using SQL, and the findings are summarized to support inventory planning, pricing, and marketing decisions.

Value	KPI
5000	Transactions
\$1.6B	Total Revenue
2023–2025	Years Covered
20	Cities

Methodology

The analysis was performed using SQL, including GROUP BY, aggregate functions, window functions (RANK()), subqueries, and Common Table Expressions (CTEs).

Before the analysis, a data quality assessment was conducted:

- No duplicate records were found, No null or blank values were detected.
- The Order Date column was converted to the DATE data type for time-based analysis.

Business Questions

- Which product generated the highest revenue?
- How loyal are the store's customers based on purchasing behavior?
- Which region and city generated the highest revenue and sales quantity?
- Which category and sub-category generated the highest sales and profit?
- Which payment mode is most frequently used by customers?
- How do monthly sales and profit change over time?
- Is there a relationship between discount and sales revenue across sub-categories?

Key Results

- Product Performance

Category Results	Sub-category Results	Metric
Home Decor (\$57M)	Sofa (15M\$)	Highest Revenue
Furniture (\$8.6M)	Sofa (2.3M\$)	Highest Profit
Furniture (15.35%) Clothing (15.34%)	Mixer Grinder (17%)	Highest Profit Margin
Groceries (47M\$)	Cricket bat (9M\$)	Lowest Revenue
Groceries(7M\$)	Dumbbells (1.2M\$)	Lowest Profit

Result: Home Decor generated the highest revenue, while Furniture achieved the strongest overall profitability, While sofa generated the highest sub- Category across revenue and profit.

- Customer Behavior

Purchase Count	Customers
3 Purchase	96.8%
6 Purchase	3.1%
9 Purchase	0.04%

Result: 96.82% of customers placed 3 orders, 3.14% placed 6 orders, and only 0.04% placed 9 orders.

- Region And City Performance

Results	Metric
North (143M\$)	Highest Revenue Region
North (21M\$)	Highest Profit Region
Bangalore (29M\$)	Highest Revenue City
Lucknow (4.5M\$)	Highest Profit City

Result: North recorded the strongest region across revenue and profit. Bangalore generated the highest revenue, while Lucknow achieved the highest profit despite similar revenue performance.

- Yearly & Monthly Sales

Year (Highest to Lowest)	Highest Month	Lowest Month
2024 (271M\$)	December	February
2025(198M\$)	May	October
2023(63M\$)	November	December

Result: 2024 generated the highest revenue and profit, making it the strongest-performing year. Since 2023 contains only three months of data, its performance should not be directly compared with 2024 and 2025. October 2025 recorded only 18 transactions, making it the most significant sales outlier in the dataset.

Key Insights

- Furniture and Home Decor are the strongest-performing categories, leading across revenue, profit, and profit margin. In contrast, Groceries and Beauty consistently underperform and require further investigation.
- Sofa emerged as the best-performing sub-category, generating the highest revenue, total profit, and sales quantity, making it the primary revenue driver of the business.
- Customer purchasing behavior is highly concentrated: 96.82% of customers made exactly three purchases, while only a small proportion progressed to six (3.14%) or nine purchases (0.04%). This indicates that highly frequent purchasers are rare and represent a valuable customer segment.
- Customer payment preferences are relatively balanced, indicating that no single payment method dominates customer transactions.
- Regional performance is relatively balanced, with the North generating the highest revenue and profit. At the city level, Guwahati leads sales activity, while product preferences vary significantly across the 20 cities, indicating localized customer demand.
- 2024 was the strongest-performing year. Since 2023 contains only three months of data, it should not be directly compared with the following years. October 2025 recorded only 18 transactions, making it the dataset's most significant outlier and a case that should be validated for data quality or operational issues.
- The discount analysis shows a weak positive relationship between discount and sales revenue, indicating that higher discounts do not consistently lead to higher sales across sub-categories.

Recommendation

- Strengthen high-performing categories: Continue investing in the Furniture category and Sofa sub-category by maintaining inventory levels and analyzing the factors behind their strong sales and profitability.
- Improve low-performing categories: Review pricing, promotions, product assortment, and customer feedback for the Beauty category to improve both revenue and profitability.
- Increase repeat purchasing frequency: Since nearly all customers stop at 3 purchases, the business should encourage progression to higher purchase levels through loyalty rewards, personalized offers, or milestone-based promotions (e.g., rewards after the 4th or 5th purchase).
- Optimize regional strategy: Maintain the strong performance of the North region while investigating the lower performance of Thiruvananthapuram to identify local market opportunities.
- Evaluate discount effectiveness: Analyze whether higher discounts generate sufficient additional sales to justify the reduction in profit margin, particularly across different sub-categories.
- Investigate the October 2025 outlier: Verify whether the unusually low 18 transactions resulted from a genuine business event or a data quality issue before making strategic decisions.